

Start–Stop MDP: A Small Numerical Example

Same next state can be reached with different rewards

State	Meaning	Actions
R	Ready state	Start, Stop
W	Work has started	–
T	Process ended	–

Full probability model

Scenario

The agent is in state R. It can choose Start or Stop. Under Start, there are two different ways to reach W, with different rewards.

Current state s	Action a	Next state s'	Reward r	$p(s', r \mid s, a)$
R	Start	W	4	0.6
R	Start	W	2	0.2
R	Start	R	0	0.2
R	Stop	T	1	1.0

$p(s', r \mid s, a)$ = probability of next state and reward together.

Step 1: Check that probabilities are valid

Scenario

For each fixed state-action pair, all outcome probabilities must add to 1.

For (R, Start):

$$\begin{aligned} p(W, 4 \mid R, \text{Start}) + p(W, 2 \mid R, \text{Start}) + p(R, 0 \mid R, \text{Start}) \\ = 0.6 + 0.2 + 0.2 = 1.0. \end{aligned}$$

For (R, Stop):

$$p(T, 1 \mid R, \text{Stop}) = 1.0.$$

So the probability model is valid for both actions.

Step 2: From $p(s', r \mid s, a)$ to $p(s' \mid s, a)$

Scenario

Compute the state-transition probability for action Start.

The reward is removed by summing over all possible rewards:

$$p(s' \mid s, a) = \sum_{r \in \mathcal{R}} p(s', r \mid s, a).$$

For action Start:

$$\begin{aligned} p(W \mid R, \text{Start}) &= p(W, 4 \mid R, \text{Start}) + p(W, 2 \mid R, \text{Start}) \\ &= 0.6 + 0.2 = 0.8. \end{aligned}$$

$$p(R \mid R, \text{Start}) = p(R, 0 \mid R, \text{Start}) = 0.2.$$

Step 3: Transition probability for Stop

Scenario

Compute the state-transition probability for action Stop.

For action Stop, there is only one possible outcome:

$$p(T \mid R, \text{Stop}) = p(T, 1 \mid R, \text{Stop}) = 1.0.$$

State-action pair	Next state	$p(s' \mid s, a)$
(R, Start)	W	0.8
(R, Start)	R	0.2
(R, Stop)	T	1.0

Step 4: Expected reward $r(s, a)$ for Start

Scenario

Compute the average immediate reward when action Start is chosen in state R.

$$r(s, a) = \mathbb{E}[R_t \mid S_{t-1} = s, A_{t-1} = a] = \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} p(s', r \mid s, a).$$

For (R, Start):

$$r(\text{R}, \text{Start}) = 4(0.6) + 2(0.2) + 0(0.2).$$

$$r(\text{R}, \text{Start}) = 2.4 + 0.4 + 0 = 2.8.$$

Step 5: Expected reward $r(s, a)$ for Stop

Scenario

Compute the average immediate reward when action Stop is chosen in state R.

For (R, Stop):

$$r(R, \text{Stop}) = 1(1.0) = 1.0.$$

Therefore:

$$r(R, \text{Start}) = 2.8, \quad r(R, \text{Stop}) = 1.0.$$

In this one-step reward comparison, Start has the larger expected immediate reward.

Step 6: Conditional expected reward $r(s, a, s')$

Scenario

Now the current state, action and next state are all fixed.

The conditional expected reward is:

$$r(s, a, s') = \mathbb{E}[R_t \mid S_{t-1} = s, A_{t-1} = a, S_t = s'].$$

It is computed by normalizing over the probability of the selected next state:

$$r(s, a, s') = \sum_{r \in \mathcal{R}} r \frac{p(s', r \mid s, a)}{p(s' \mid s, a)}, \quad \text{where } p(s' \mid s, a) > 0.$$

Step 7: $r(R, \text{Start}, W)$

Scenario

Action Start is chosen, and the next state is known to be W. There are two reward possibilities: 4 and 2.

From the table:

$$p(W, 4 \mid R, \text{Start}) = 0.6, \quad p(W, 2 \mid R, \text{Start}) = 0.2,$$

$$p(W \mid R, \text{Start}) = 0.8.$$

Therefore:

$$\begin{aligned} r(R, \text{Start}, W) &= 4 \frac{0.6}{0.8} + 2 \frac{0.2}{0.8}. \\ &= 4(0.75) + 2(0.25) = 3.0 + 0.5 = 3.5. \end{aligned}$$

Step 8: Other conditional rewards

Scenario

Compute the remaining conditional rewards from the same table.

For action Start and next state R:

$$r(R, \text{Start}, R) = 0 \times \frac{0.2}{0.2} = 0.$$

For action Stop and next state T:

$$r(R, \text{Stop}, T) = 1 \times \frac{1.0}{1.0} = 1.$$

These values are conditional on the particular next state that has occurred.

Deterministic special case

Scenario

Now assume both actions have exactly one possible outcome.

Current state s	Action a	Next state s'	Reward r	$p(s', r \mid s, a)$
R	Start	W	4	1.0
R	Stop	T	1	1.0

$$p(W \mid R, \text{Start}) = 1, \quad r(R, \text{Start}) = 4.$$

$$p(T \mid R, \text{Stop}) = 1, \quad r(R, \text{Stop}) = 1.$$

Final summary

Scenario

The same formulas handle both stochastic and deterministic actions.

Quantity	Meaning in this example
$p(s', r \mid s, a)$	Full probability of next state and reward together.
$p(s' \mid s, a)$	Transition probability after summing out reward.
$r(s, a)$	Average immediate reward for choosing an action in a state.
$r(s, a, s')$	Average immediate reward when the next state is also fixed.

Two rows can have the same s' but different rewards.